MIN-JONG BONG

Ph.D. Candidate in Inorganic Chemistry · Korea University · Advisor: Prof. Ho-Jin Son

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RESEARCH INTEREST

Ph.D. candidate in Inorganic Chemistry at Korea University (expected February 2027), specializing in the design of single-molecule Mn(I) catalysts for selective CO_2 reduction. Research centers on secondary coordination sphere engineering of well-defined molecular complexes — inspired by formate dehydrogenase enzymes — integrating experimental (operando FTIR, CV, XRD) and computational (DFT; ORCA) approaches with Python-based automation of computational workflows. Key results include a TON of ~300 with >94% formate selectivity, published in Chemical Science (2026, first author).

Keywords: Organometallic Synthesis, CO_2 Reduction, Photocatalysis, Secondary Coordination Sphere, Operando FTIR, DFT

EDUCATION

Ph.D. candidate, Inorganic Chemistry — Mar 2024 – Present
Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

M.S., Inorganic Chemistry — Mar 2022 – Feb 2024
Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

B.S., Advanced Materials Chemistry — Mar 2018 – Feb 2022
Korea University (Sejong)

PUBLICATIONS

[1] **Secondary Sphere Impact on Organometallic Catalysts in Photochemical CO_2 Reduction: A Comparative Study of Tethering Ligand Lengths with Secondary Coordination**
Min-Jong Bong, Wonjung Lee, Junhyeok Seo*, Ho-Jin Son*
In preparation **first author**

[2] **Unveiling Key Factors Governing the Catalytic Performance of Half-Metallocene Complexes for Photochemical CO_2 Reduction and Hydrogen Production: Relativistic Effects and Charge Delocalization**
Daehan Lee†, **Min-Jong Bong**†, Seunghwan Cha, Sangheon Jeong, Jung-Min Ji, Won-Jo Jung, Nady A. Fathy, Ho-Jin Son*
Inorganic Chemistry Frontiers · Under revision · *Royal Society of Chemistry* **co-first author †**

[3] **A secondary-sphere proton channel accelerating metal–hydride formation in Mn(I) catalysts for selective CO_2-to-formate conversion**
Min-Jong Bong†, Wonjung Lee†, Daehan Lee, Hyunuk Kim, Junhyeok Seo*, Ho-Jin Son*
Chemical Science · 2026 · *Royal Society of Chemistry* · DOI: 10.1039/D5SC09412G **first author**

[4] **Efficient and Durable Photochemical CO_2 Reduction by TiO_2-Immobilized Metal Porphyrin Catalysts**
Hyeongu Kang, Daehan Lee, Sangheon Jeong, Seunghwan Cha, **Min-Jong Bong**, Myung Jae Lee, Won-Jo Jung, Ho-Jin Son*
ChemSusChem, 19, e202501889 · 2026 · *Wiley-VCH / Chemistry Europe* · DOI: 10.1002/cssc.202501889

[5] **Organic phosphorescence in Pt(II)-complexes linked to organic chromophores for blue-emitting organic light-emitting diodes**
Seong Woon Jeong†, Hyung Joo Lee†, Bum June Park, **Min-Jong Bong**, Daehan Lee, Taekyung Kim*, Chul Hoon Kim*, Ho-Jin Son*
Communications Chemistry, 8, 140 · 2025 · *Nature Portfolio* · DOI: 10.1038/s42004-025-01533-y

RESEARCH PROJECTS

Tether Length Modulation of Secondary Coordination Layer

2024 – Present

Systematic variation of −OH/−OMe pendant length at 6,6′-bipy positions of Mn(I) complexes. Best performer Mn-DiMeOMe: TON 368, formate selectivity 97.8%.
Photocatalysis · CV · Operando FTIR · DFT · XRD

Proton Relay Engineering in Mn(I) Complexes

2022 – 2025

Ethylene-bridged pendants mimic formate dehydrogenase. Mn-bpydiOMe: TON ~300, formate selectivity >94%. Published in Chem. Sci. 2026.
CO_2 Reduction · Operando FTIR · DFT · Mechanistic Study

SKILLS & METHODS

Technical: Organic & organometallic synthesis, Glovebox & Schlenk techniques, Air-sensitive compound handling, Photocatalysis, Sublimation purification, 3D Graphic Design (Adobe Dimension)

Instrumental: Operando FTIR, Cyclic Voltammetry, UV-Vis Spectroscopy, Quantum Yield Measurement, Single-crystal XRD, Structure refinement (SHELXTL/SHELXL), NMR, ESI-MS, GC-MS, GC-TCD, HPLC

Computational: DFT (ORCA, Gaussian), CPCM solvation, Reaction pathway mapping, Python (computational data automation & visualization), AI-assisted computational workflow (Claude Code)

TEACHING & MENTORING

General Chemistry Laboratory 1 Lecturer — Spring 2026
Serving as course instructor for general chemistry laboratory.

Inorganic Chemistry Laboratory Lecturer — Fall 2026
Serving as course instructor for inorganic chemistry laboratory.

CURT Undergraduate Research Mentoring — Feb – Jul 2024
Mentored undergraduates in research design and execution.

General Chemistry Laboratory 1 TA — Spring 2022, 2023, 2024, 2025
Supervised lab sessions for general chemistry.

General Chemistry Laboratory 2 TA — Fall 2022, 2024
Supervised lab sessions for general chemistry.

Inorganic Chemistry Laboratory TA — Fall 2023, 2024, 2025
Instructed inorganic synthesis experiments.

HONORS & AWARDS

Excellence in Poster Presentation Award 2025

Inorganic Chemistry Division, Korean Chemical Society

Second Coordination Sphere Engineering for Efficient and Selective CO_2-to-Formate Conversion of Mn(I) Bipyridyl Complexes: Steric and Brønsted Acidity/Basicity Influence of Alcohol and Alkyloxy Pendants and Its Mechanistic Investigation

Excellent Research Award 2021

CURT Research Program, Korea University

A Hybrid Ru/TiO_2 Catalyst for Photocatalytic CO_2 to CO/Formate Conversion